MIL-C-83723/12B(USAF)
1 OCT 1975
SUPERSEDING
MIL-C-83723/12A(USAF)
19 April 1971

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING), (SERIES I, CLASS H)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Ⓑ The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-26482.

Ⓑ INACTIVE. SUPERSEDED BY MS3449 of MIL-C-26482.

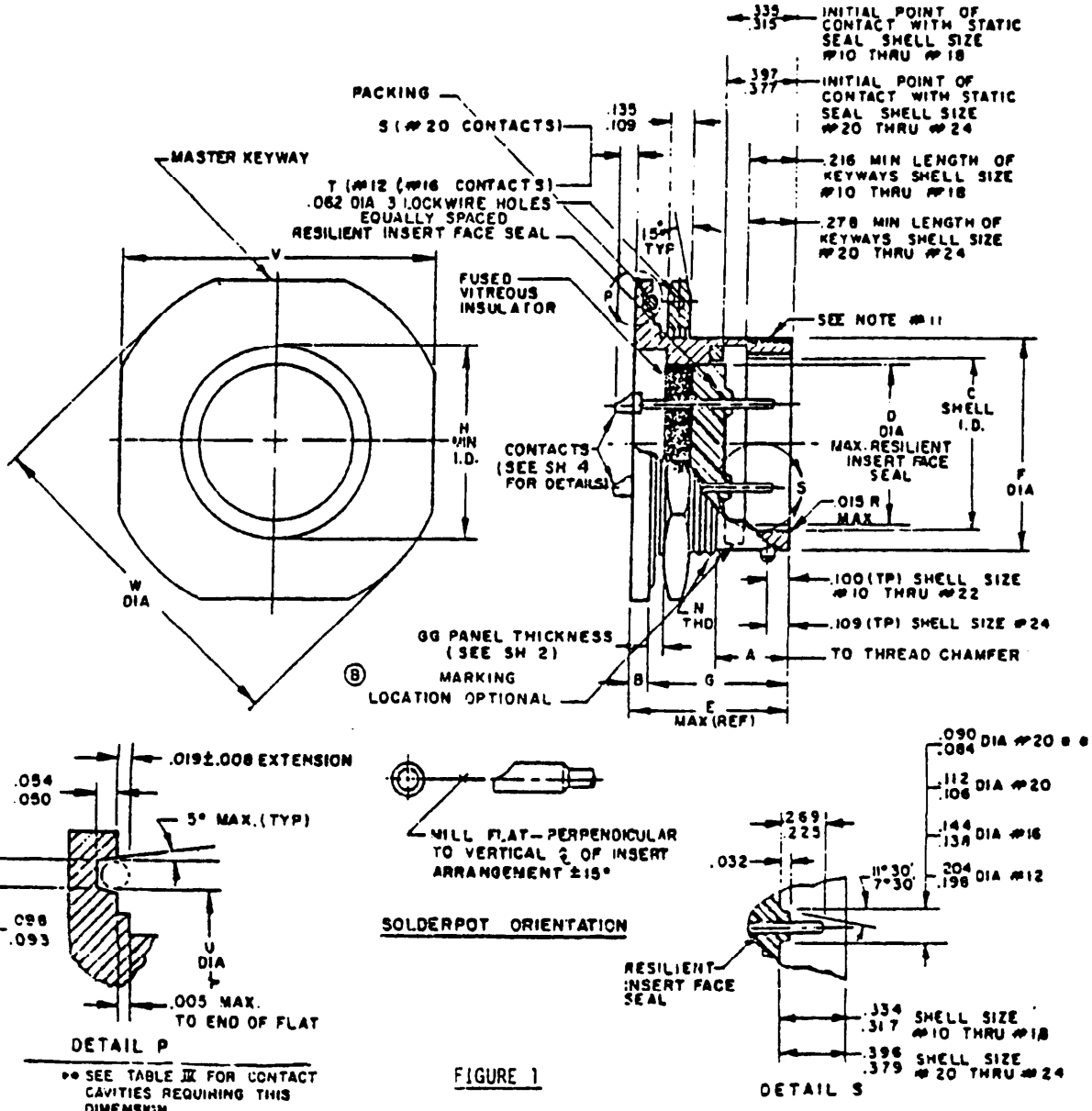


FIGURE 1

Ⓑ denotes changes

FSC 5935

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TABLE I

CONNECTOR DIMENSIONS

SHELL SIZE	A	B	C	D	E	F	G	H	I	N	S	T	U	V	W
10	.378 .358	.113 .097	.495 .489	.402	.820	.591 .585	.707 .691	.515		11/16-24 UNEF-2A	.134 .074	.204 .144	.734	1.078 1.047	1.203 1.172
12	.378 .358	.113 .097	.612 .606	.516	.820	.751 .745	.707 .691	.630		7/8-20 UNEF-2A	.134 .074	.204 .144	.921	1.266 1.235	1.391 1.360
14	.378 .358	.113 .097	.737 .731	.641	.820	.876 .870	.707 .691	.755		1-20 UNEF-2A	.134 .074	.204 .144	1.046	1.391 1.360	1.516 1.485
16	.378 .358	.113 .097	.862 .856	.766	.820	1.001 .995	.707 .691	.880		1-1/8-18 UNEF-2A	.134 .074	.204 .144	1.171	1.516 1.485	1.641 1.610
18	.378 .358	.113 .097	.967 .961	.855	.820	1.126 1.120	.707 .691	.980		1-1/4-18 UNEF-2A	.134 .074	.204 .144	1.296	1.641 1.610	1.766 1.735
20	.378 .358	.148 .128	1.092 1.086	.980	.920	1.251 1.245	.772 .754	1.105		1-3/8-18 UNEF-2A	.099 .039	.169 .109	1.484	1.828 1.797	1.954 1.923
22	.378 .358	.148 .128	1.217 1.211	1.105	.920	1.376 1.370	.772 .754	1.230		1-1/2-18 UNEF-2A	.099 .039	.169 .109	1.609	1.954 1.923	2.078 2.047
24	.405 .385	.148 .128	1.342 1.336	1.229	.951	1.501 1.495	.803 .785	1.385		1-5/8-18 UNEF-2A	.099 .009	.169 .079	1.734	2.078 2.047	2.203 2.172

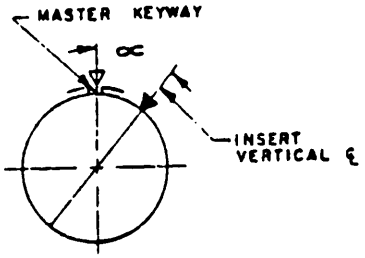
SIZE	X	Y	Z	AA	EE	FF	GG
10	.650	.545 .534	.680 .664	.892	.661	.697	.187 .062
12	.813	.694 .683	.859 .843	1.079	.824	.895	.187 .062
14	.937	.819 .808	.984 .968	1.205	.948	1.010	.187 .062
16	1.061	.944 .933	1.108 1.092	1.329	1.072	1.135	.187 .062
18	1.186	1.044 1.033	1.233 1.217	1.455	1.197	1.260	.187 .062
20	1.311	1.169 1.158	1.358 1.342	1.579	1.322	1.385	.250 .062
22	1.436	1.294 1.283	1.483 1.467	1.705	1.447	1.510	.250 .062
24	1.561	1.419 1.408	1.610 1.594	1.829	1.572	1.635	.250 .062

TABLE II

CONTACT DIMENSIONS

CONT SIZE	BB	CC	DD	HH	KK
20	.041 .039	.088 .061	.048 .042	.188 .109	.110 .068
16	.0635 .0615	.116 .096	.075 .069	.251 .172	.141 .109
12	.095 .093	.150 .130	.122 .112	.251 .172	.141 .109

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(AS VIEWED FROM ENGAGING END)

INSERT POSITION

▽ INDICATES ℄ OF SHELL

FIGURE 2

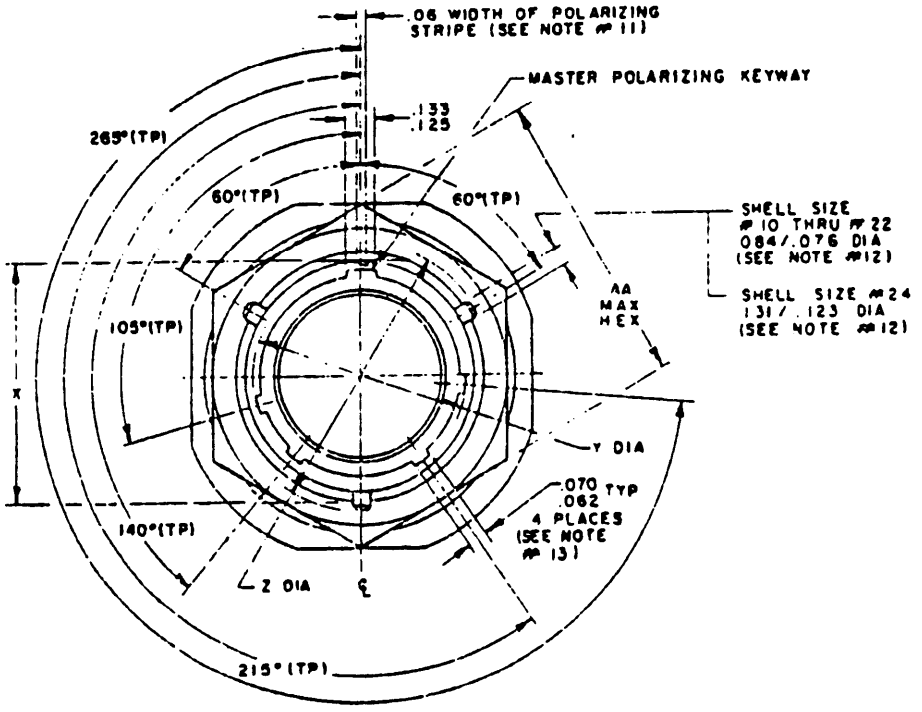
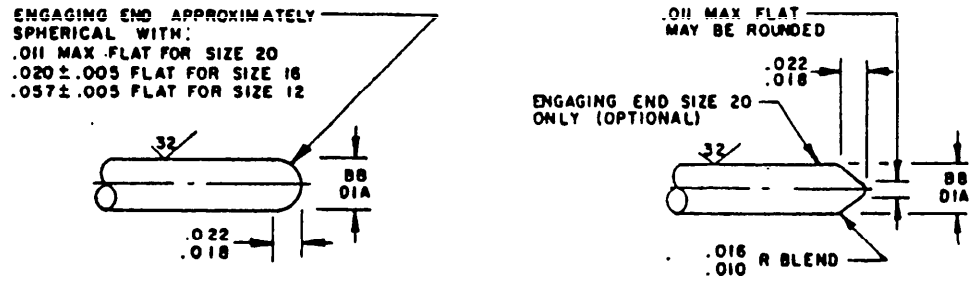


FIGURE 3

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ENGAGING END OF CONTACTS



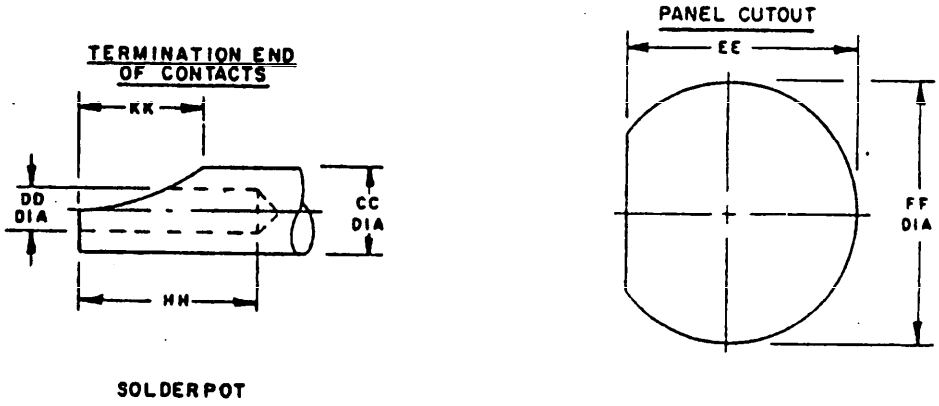


FIGURE 4

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TABLE III. Insert arrangements requiring reduced diameters for raised seal barriers on outer row of contact cavities as indicated.

Shell	Insert arrangements	Contact cavities
12	10	C and G
14	12	A,B,C,D,E,F,G, and H
14	18	A,C,E,G,J, and L
14	19	B,D,F,H,K, and M
16	26	A,B,C,D,E,F,G,H,J, K,L,M,N,P, and R
18	32	A,B,C,D,E,F,G,H,J, K,L,M,N,P,R,S, and T
22	41	A,B,C,D,E,F,G,H,J, K,L,M,N,P,R,S,T,U, V,W,X, and Y

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NOTES: (NOTES 1 THRU 6 APPLICABLE TO CONTACTS ONLY).

1. CONTACT FINISH: GOLD PLATE .000050 INCH MIN. IN ACCORDANCE WITH MIL-G-45204. A SUITABLE UNDERPLATING SHALL BE USED. FINISH IS APPLIED AFTER SEALING.

2. DIMENSIONS IN INCHES. DIMENSIONS APPLY AFTER PLATING.

3. 63√ALL OVER PER ANSI B46.1-62 UNLESS OTHERWISE SPECIFIED

4. REMOVE ALL BURRS AND SHARP EDGES .003 TO .005 R.

5. DIAMETERS TO BE CONCENTRIC WITHIN .004 TIR.

6. DIAMETERS IN THE SAME CROSS-SECTIONAL PLANE TO BE CONCENTRIC WITH EACH OTHER WITHIN .003 TIR.

7. FOR ALTERNATE POLARIZING POSITIONS, THE CONNECTOR INSERT IS ROTATED IN RESPECT TO THE SHELL.

8. IN "NORMAL POSITION" (POSITION "N") THE INSERT VERTICAL CENTERLINE COINCIDES WITH THE CENTERLINE OF THE MASTER KEYWAY OF THE SHELL.

9. IN "ALTERNATE POSITIONS" (POSITIONS "W", "X", "Y" AND "Z") THE PIN INSERT IS ROTATED α DEGREES CLOCKWISE RELATIVE TO $\mathcal{E}$ OF THE MASTER KEYWAY OF SHELL. SEE REQUIREMENT 4.

10. POLARIZING STRIPE SHALL BE COLOR CODED BLUE.

11. THREE (3) PROJECTIONS LOCATED WITHIN .005 R (TP) AT (MMC) RELATIVE TO "C" DIA (MMC) AND SHELL FRONT, AND WITHIN .010 EITHER SIDE OF (TP) AT (MMC) RELATIVE TO MASTER POLARIZING KEYWAY.

12. FOUR KEYWAYS (MMC) AND INSERT SHALL BE LOCATED WITHIN .0035 EITHER SIDE OF (TP) RELATIVE TO MASTER KEYWAY (MMC) AND SHELL I.D. (MMC).

13. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCE SHALL BE .XX ±.010 AND .XXX ± .005; ANGULAR TOLERANCE X° ±1 -0' AND X°X' ±30'.

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TABLE IV WIRE DATA

CONTACT SIZE	WIRE SIZE AWG	INSULATION O.D. IN INCHES / WIRE SPECIFICATION 3/			
		MIL-W-22759/4	MIL-W-16878/4	MIL-W-81381/-	
				LIGHT WT -/1	MEDIUM WT -/3
20	24	-	.040/.048 1/	.037/.043	.047/.054
	22	.071/.077	.046/.054	.043/.050	.053/.060
	20	.079/.085 2/	.054/.062	.051/.058	.061/.068
16	20	.079/.085	.054/.062 1/	.051/.058	.061/.068
	18	.092/.098	.064/.074	.060/.067	.071/.079
	16	.099/.107 2/	.073/.087	.066/.074	.079/.087
12	14	.112/.120 1/	.087/.101	.080/.089	.091/.099
	12	.129/.137 2/	.106/.120	.098/.108	.107/.117

1/ WIRE TO BE USED FOR CONNECTOR QUALIFICATION.

2/ WIRE TO BE USED FOR CONTACT QUALIFICATION.

3/ LISTED WIRE SPECIFICATIONS ARE NOT ALL INCLUSIVE. WIRE WITH SMOOTH EXTERIOR SURFACE, WHICH FALLS WITHIN THE SPECIFIED RANGE AND MEETS THE DESIGNERS CRITERIA, MAY BE USED IN APPLICATION.

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REQUIREMENTS:

Dimensions and configuration: See figures.

This connector mates with MIL-C-83723/13(USAF).

(B) For insert arrangements see MIL-STD-1669.

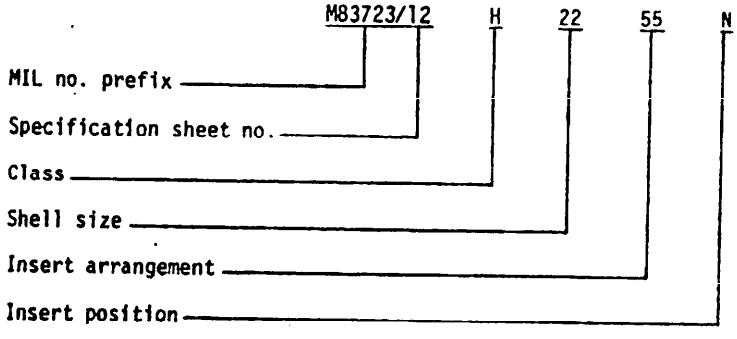
"O" ring shall be furnished with connector.

Shell material: Cold rolled steel per QQ-S-637.

Wire: See table IV.

(B) Connector identification marking shall be in accordance with MIL-C-26482. Example: MS3449L----, M83723/12R----.

Part number example:



Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 70, 71, 80, 82, 84
DSA - ES

Preparing activity:
Air Force - 85

(Project 5935-F964-12)

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